

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 7: Coordinate Geometry

Previous year board questions, topic-wise: 8 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The distance formula

1. A circle has its centre at $(2a, a - 7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a . **CBSE 2025, 2 marks**
2. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD . (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**
3. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x . (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**

2 The section formula

4. Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? **CBSE 2025, 1 mark**
5. In what ratio does the y -axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. **CBSE 2025, 3 marks**
6. A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**
7. The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. **CBSE 2024, 3 marks**
8. $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P , Q , R and S are the mid-points of AB , BC , CD and DA . Show that the diagonals of $PQRS$ bisect each other. **CBSE 2024, 3 marks**

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.